One-Day-Ahead 1% VaR Forecasting on QQQ Daily Log Returns

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1 Introduction

This report studies one-day-ahead forecasting of the 1% Value-at-Risk (VaR) for QQQ daily log returns. Let r_t denote the QQQ log return on day t. The forecasting target is the conditional 1% left-tail quantile of r_{t+1} given information available at the close of day t. In practical terms, the question is: how negative could tomorrow's return be, with 1% probability, conditional on current market information?

The project is organized as a rolling out-of-sample forecasting exercise on QQQ returns, with VIX used as an exogenous volatility regressor in one comparison model. The emphasis is not in-sample fit but out-of-sample tail-risk forecasting under a realistic workflow: estimate each model using data available through time t, generate a one-day-ahead VaR forecast for $t+1$, and then record whether the realized return violates the forecast.

The current repository outputs show a clear pattern. Methods that combine time-varying volatility with empirical tail calibration perform best. In particular, GJR-FHS is the strongest overall model in the saved results, followed closely by GARCH-FHS. Student-t empirical-tail variants also perform well, while Gaussian VaR methods are too thin-tailed for QQQ downside risk. Historical Simulation remains a credible benchmark. The updated package-based GARCHX(VIX) model is methodologically cleaner than the earlier custom implementation, but in the saved outputs it is still weaker than the best empirical-tail alternatives.

2 Historical Background and Motivation

VaR emerged as a standard market-risk summary in the 1990s as financial institutions sought a single, interpretable measure of downside exposure over a fixed horizon and confidence level. A major practical milestone was the *RiskMetrics* framework released by J.P. Morgan in 1994, which popularized volatility-based VaR and helped standardize daily market-risk reporting. Regulatory adoption followed quickly: the 1996 Basel market-risk amendment allowed banks to use internal VaR models for capital calculations, making VaR central to both practice and regulation.

The statistical foundations of VaR forecasting are closely tied to conditional heteroskedasticity models. Engle's ARCH model and Bollerslev's GARCH model provided a formal framework for volatility clustering, a central empirical feature of financial returns. These models made it possible to update conditional variance dynamically and map that variance into tail-risk forecasts.

Once VaR models became widespread, model validation became equally important. Kupiec (1995) proposed the unconditional coverage test, which checks whether the observed fraction of VaR violations matches the nominal target probability. Christoffersen (1998) extended this framework by testing whether violations are independent over time, since clustered breaches indicate that a model is not tracking changing risk conditions well even if the overall hit rate looks acceptable.

A different strand of the literature moved away from modeling the full return distribution first and instead modeled the quantile directly. Engle and Manganelli's CAViaR framework is the classic example. Its key idea is that the VaR itself can follow a dynamic process estimated by quantile regression methods, avoiding some strong distributional assumptions but requiring careful optimization and adequacy checking.

These developments motivate the current project. The central comparison is not only between different volatility recursions, but also between different ways of translating information into a left-tail quantile: Gaussian parametric tails, Student-t tails, empirical residual tails, direct quantile dynamics, and an exogenous-volatility benchmark using VIX.

3 Purpose Statement and Mathematical Development

3.1 Purpose of VaR forecasting

For a one-day-ahead VaR forecast at level $\alpha = 0.01$, the target is

$$\Pr(r_{t+1} < \mathrm{VaR}_{t+1}(\alpha) \mid \mathcal{F}_t) = \alpha,$$

where $\mathcal{F}_t$ is the information set available at forecast origin t. In this project,

$$r_t = \log\left(\frac{P_t}{P_{t-1}}\right),$$

where P_t is the adjusted closing price of QQQ.

For most models in the repository, returns follow a conditional location-scale form,

$$r_{t+1} = \mu_{t+1} + \sigma_{t+1} z_{t+1},$$

which implies

$$\mathrm{VaR}_{t+1}(\alpha) = \mu_{t+1} + \sigma_{t+1} q_\alpha,$$

where q_α is either a parametric lower-tail quantile or an empirical residual quantile.

3.2 Data setting and rolling design

The project uses QQQ adjusted close data and VIX close data beginning on 2015-01-01. QQQ prices are converted to daily log returns. VIX enters only through the GARCHX comparison model. The saved outputs indicate:

- a modeled sample beginning in early 2015;
- a 70% initial training fraction;
- an out-of-sample period of 848 one-day forecasts per model;
- expected violations equal to 8.48 at the 1% VaR level.

At each out-of-sample date, the workflow is:

1. estimate the model using all data through time t;
2. forecast one-day-ahead VaR for time $t+1$;
3. compare the realized return to the forecast;
4. record the violation indicator

$$I_{t+1} = \mathbf{1}(r_{t+1} < \widehat{\mathrm{VaR}}_{t+1});$$

5. backtest the violation sequence.

This rolling design is essential because VaR is a forecasting problem. A model that looks good in-sample but fails on genuine rolling forecasts is not useful for risk management.

3.3 Model framework used in the project

3.3.1 GARCH(1,1)

The standard GARCH(1,1) model specifies

$$\varepsilon_t = r_t - \mu,$$

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2.$$

In the main project runner, the saved baseline uses Student-t innovations, so one-step VaR is

$$\widehat{\mathrm{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} q_\alpha^{(t)},$$

where $q_\alpha^{(t)}$ is the fitted Student-t lower-tail quantile.

3.3.2 GJR-GARCH

To allow leverage effects, GJR-GARCH uses

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \gamma\varepsilon_{t-1}^2 I(\varepsilon_{t-1} < 0) + \beta\sigma_{t-1}^2.$$

Negative shocks therefore have an extra effect on future volatility when $\gamma > 0$, which is especially relevant for equity returns.

3.3.3 Historical Simulation

Historical Simulation is fully nonparametric. It uses the empirical lower-tail quantile of the most recent W returns:

$$\widehat{\mathrm{VaR}}_{t+1}(\alpha) = \mathrm{Quantile}_\alpha(r_{t-W+1}, \dots, r_t).$$

In the repository, the default window is $W = 250$. This method is transparent and easy to interpret, but it can adjust slowly when volatility changes abruptly.

3.3.4 Filtered Historical Simulation (FHS)

FHS first estimates a volatility model, then replaces a parametric tail quantile with the empirical quantile of standardized residuals. Its one-step forecast is

$$\widehat{\mathrm{VaR}}_{t+1}(\alpha) = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1}\hat{q}_{\alpha,\mathrm{emp}}.$$

The project includes both Gaussian-based and Student-t-based FHS variants, including GARCH-FHS, GJR-FHS, GARCH-t-FHS, and GJR-t-FHS. These are semi-parametric methods: the volatility dynamics are modeled parametrically, but the tail quantile is learned from the empirical residual distribution.

3.3.5 Residual bootstrap VaR

The residual bootstrap approach resamples standardized residuals from a fitted GARCH-type model, simulates one-step-ahead returns, and then takes the empirical 1% quantile of those simulated returns. For one-step forecasting, this is mechanically very close to FHS, which helps explain why bootstrap and FHS forecasts are similar in the saved outputs.

3.3.6 CAViaR-SAV

CAViaR models the quantile directly rather than through a volatility recursion. The saved implementation uses the Symmetric Absolute Value (SAV) form

$$v_t = \beta_0 + \beta_1 v_{t-1} + \beta_2 |r_{t-1}|,$$

followed by

$$q_t = -v_t.$$

The one-step VaR forecast is the next recursively updated quantile. Parameters are estimated by minimizing the check-loss function

$$L(\beta) = \sum_t \rho_\alpha(r_t - q_t(\beta)),$$

where

$$\rho_\alpha(u) = u(\alpha - I(u < 0)).$$

This is faithful to the direct-quantile spirit of Engle and Manganelli, but the current implementation remains simplified: only the SAV variant is used, the optimization is limited, and the Dynamic Quantile test is not implemented.

3.3.7 GARCHX(VIX)

The updated exogenous-volatility benchmark is implemented in `GARCHX.r` using the R `garchx` package. In each rolling window, returns are demeaned and VIX is standardized. A simple representation is

$$y_t = r_t - \mu,$$

$$\sigma_t^2 = \omega + \alpha y_{t-1}^2 + \beta \sigma_{t-1}^2 + \delta x_t,$$

where x_t is standardized VIX. The one-step VaR forecast is then formed as

$$\widehat{\mathrm{VaR}}_{t+1}(\alpha) = \mu + \hat{\sigma}_{t+1|t} \Phi^{-1}(\alpha).$$

This is best described as a package-based Gaussian-QML GARCHX benchmark. It is methodologically cleaner than the earlier custom implementation, but its tail mapping remains Gaussian.

4 Representative Data Example and Code

The representative data example is the actual project dataset: QQQ daily log returns, with VIX included for the GARCHX comparison. The saved pipeline is built around the following commands:

```
python Data_Scraping/fetch_data.py --start 2015-01-01 --outdir data
python run_var_project.py --data data/model_data.csv --outdir outputs --alpha 0.01 --initial-tr
python run_var_ext.py --data data/model_data.csv --outdir outputs --alpha 0.01 --initial-train
python run_var_riskmetrics.py --data data/model_data.csv --outdir outputs/riskmetrics --alpha 0
Rscript GARCHX.r
```

The following project-style code sketch captures the core rolling forecasting logic:

```python
import pandas as pd

df = pd.read_csv("data/model_data.csv", parse_dates=["Date"])
df = df[["Date", "log_ret", "vix_close"]].dropna().sort_values("Date").reset_index(drop=True)

alpha = 0.01
initial_train = 0.7
n = len(df)
train_end0 = int(n * initial_train)

rows = []
for t in range(train_end0, n):
    train = df.iloc[:t]
    test_row = df.iloc[t]

    r_train = train["log_ret"].to_numpy()
    realized = float(test_row["log_ret"])

    var_t = fitted_model_forecast(r_train, alpha=alpha)
    hit_t = int(realized < var_t)

    rows.append({
        "Date": test_row["Date"],
        "VaR": var_t,
        "Return": realized,
        "Violation": hit_t,
    })
```

The key feature is that each forecast is generated only from information available at that date, which avoids look-ahead bias.

5 Empirical Results

5.1 Main backtest comparison

The current saved outputs combine backtests from the main runner, extension runner, RiskMetrics runner, and package-based GARCHX benchmark. Table 1 reports the integrated one-day-ahead 1% VaR comparison.

Model	Violations	Violation Rate	Kupiec p-value	Christoffersen p-value	Conditional Coverage p-value
GJR-FHS	8	0.94%	0.8672	0.6961	0.9137
GARCH-FHS	7	0.83%	0.5984	0.7327	0.8212

Model	Violations	Violation Rate	Kupiec p-value	Christoffersen p-value	Conditional Coverage p-value
GARCH-t-FHS	6	0.71%	0.3664	0.7698	0.6371
GJR-t-FHS	6	0.71%	0.3664	0.7698	0.6371
GARCH-t-Bootstrap	6	0.71%	0.3664	0.7698	0.6371
GARCH-Bootstrap	5	0.59%	0.1934	0.8075	0.4166
Historical-Simulation	12	1.42%	0.2528	0.1569	0.1910
GARCH(1,1)	13	1.53%	0.1480	0.1890	0.1482
GJR-GARCH	15	1.77%	0.0424	0.4621	0.0972
GARCHX(VIX) [garchx pkg]	15	1.77%	0.0424	0.2621	0.0679
CAViaR-SAV	6	0.71%	0.3664	0.0297	0.0625
RiskMetrics ($\lambda = 0.94$)	18	2.12%	0.0043	0.3925	0.0117
GARCH-Gaussian	22	2.59%	0.0001	0.5970	0.0005

Several conclusions follow immediately.

First, the best models are empirical-tail methods. GJR-FHS is the strongest overall performer: its violation rate is closest to the 1% target and it also passes the independence and conditional coverage tests comfortably. GARCH-FHS is a very close second. Student-*t* empirical-tail variants are also strong, though slightly conservative.

Second, Gaussian tail assumptions perform poorly for this dataset. GARCH-Gaussian and RiskMetrics both generate too many violations, indicating that their left tails are too light for QQQ downside risk.

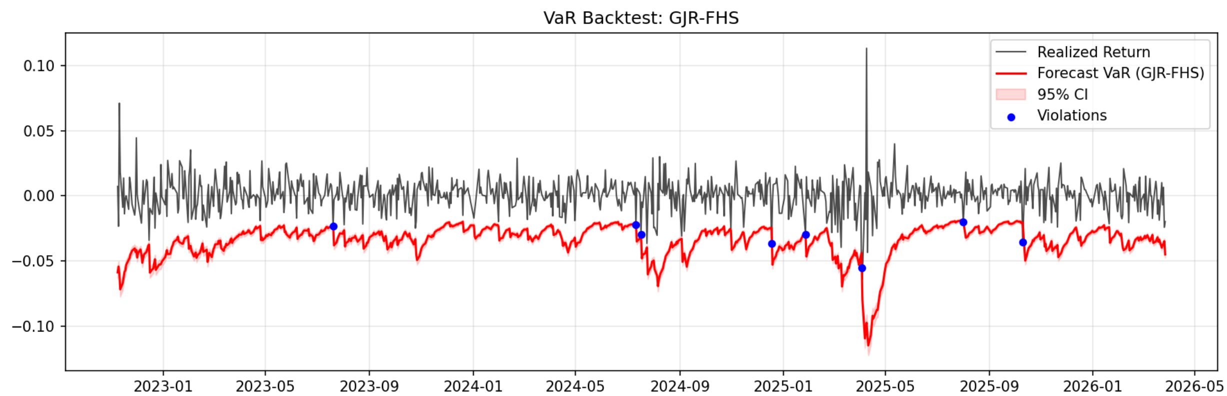
Third, Historical Simulation remains a useful benchmark. It is slower-moving and somewhat less adaptive than the best FHS models, but it still delivers a respectable backtesting profile.

Fourth, the updated package-based GARCHX(VIX) benchmark is easier to justify methodologically than the earlier custom implementation, yet its saved results still underperform the strongest empirical-tail models. In this project, incorporating VIX into the variance recursion does not by itself produce superior one-day 1% VaR forecasts.

Finally, CAViaR-SAV deserves a more careful interpretation than before. Its hit rate is conservative and acceptable, but its Christoffersen p-value indicates clustering in the violation sequence. That means the model is not fully tracking changing risk dynamics, even though its total number of violations looks reasonable.

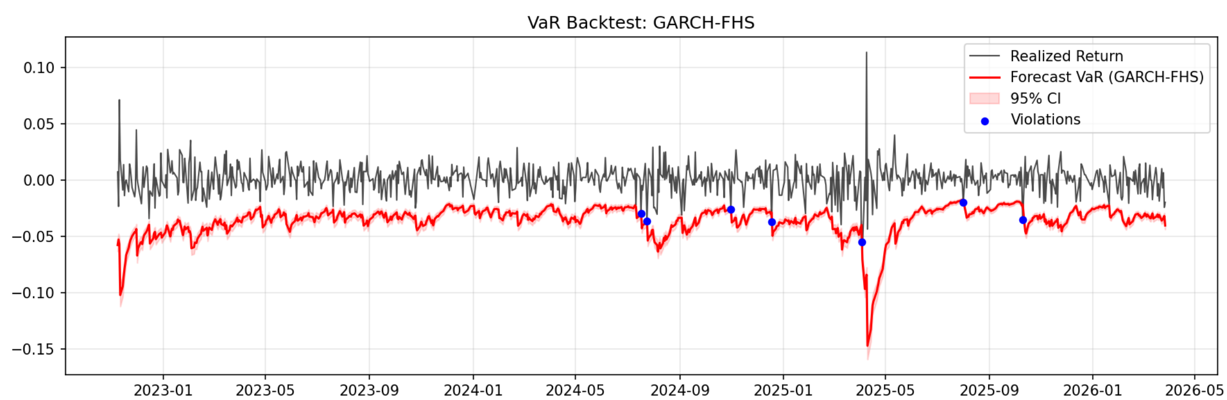
5.2 Plot-based interpretation

5.2.1 Best overall model: GJR-FHS



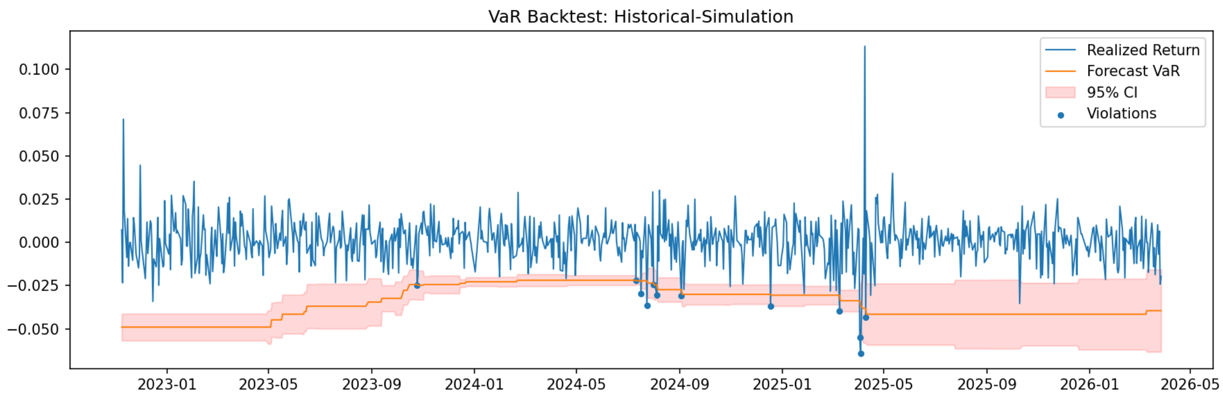
The GJR-FHS VaR line moves decisively downward during stress periods without becoming excessively conservative. This visual behavior matches the strong backtesting evidence.

5.2.2 Close second: GARCH-FHS



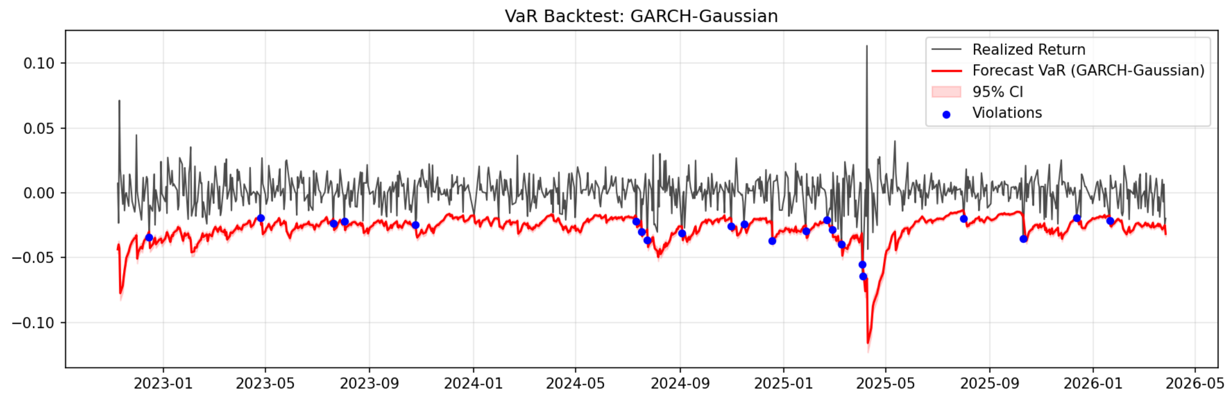
GARCH-FHS behaves similarly, with empirical tail calibration doing much of the work. It is slightly less flexible than GJR-FHS on asymmetry, but still very strong overall.

5.2.3 Benchmark comparison: Historical Simulation



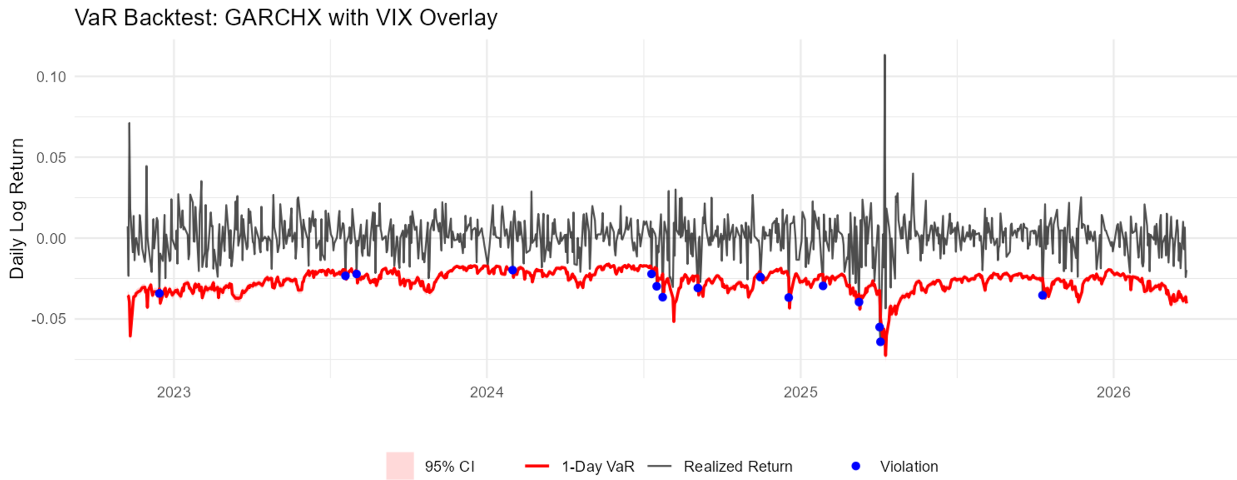
Historical Simulation is visibly more stepwise and slower-moving, but it remains a respectable benchmark because it does not rely on a strong distributional assumption.

5.2.4 Weak Gaussian benchmark



The Gaussian GARCH VaR line stays too close to zero during volatile periods. The graphical evidence and the backtests tell the same story: the Gaussian tail is too thin.

5.2.5 Package-based GARCHX benchmark



The package-based GARCHX forecast path is coherent and methodologically cleaner than the earlier custom implementation, but it still does not beat the best FHS models in calibration.

6 Handling Uncertainty in VaR Forecasting

A point VaR forecast is not the full story. VaR itself is an estimated conditional quantile, so it comes with uncertainty arising from parameter estimation, tail estimation, finite sample size, and model choice. For that reason, the updated presentation's uncertainty slides and the saved forecast files include 95% interval bounds for one-day-ahead VaR through `lower_bound` and `upper_bound` columns.

Conceptually, for a quantile estimator $\hat{q}_\alpha$, an asymptotic confidence interval takes the form

$$\hat{q}_\alpha \pm 1.96\,\mathrm{SE}(\hat{q}_\alpha),$$

with

$$\mathrm{SE}(\hat{q}_\alpha) = \sqrt{\frac{\alpha(1-\alpha)}{n\,f(\hat{q}_\alpha)^2}},$$

where $f(\hat{q}_\alpha)$ is the density at the target quantile. For parametric VaR models, this interval is carried through the location-scale mapping

$$\mathrm{VaR}_{t+1}(\alpha) = \mu_{t+1} + \sigma_{t+1} q_\alpha.$$

In the repository, the interval construction differs across model classes, consistent with slides 27–29 of the updated presentation:

- **Parametric Gaussian or Student-t models** use an asymptotic quantile-standard-error idea mapped through $\mu + \sigma q_\alpha$.
- **FHS models** combine a parametric volatility forecast with an empirical residual-tail quantile, so their VaR interval reflects uncertainty in that empirical tail estimate.
- **Bootstrap models** form intervals directly from simulated one-step returns rather than from an analytic residual-quantile formula.
- **CAViaR-SAV** uses a centered empirical proxy interval: the code shifts a recent return window so that its tail quantile matches the CAViaR point forecast, then applies a sample-quantile interval to that shifted sample.

6.1 Evidence from the saved forecast files

Using the current `*_forecasts.csv` outputs, the interval width

$$\text{width}_t = \text{upper\_bound}_t - \text{lower\_bound}_t$$

provides a practical way to compare uncertainty across models. Because all VaR forecasts are negative, a wider interval means less precision about the location of the next-day tail quantile, while a narrower interval means greater precision or, in some cases, a mechanically tighter construction.

Table 2 summarizes the average 95% VaR interval width across the saved one-day forecast files.

Model	Mean interval width
CAViaR-SAV	0.02064
Historical-Simulation	0.02064
GARCH-t-FHS	0.00783
GARCH-FHS	0.00688
GARCH(1,1)	0.00677
GJR-GARCH	0.00637
GJR-t-FHS	0.00586
GJR-FHS	0.00436
GARCH-Bootstrap	0.00303
GARCHX(VIX) [garchx pkg]	0.00353
GARCH-t-Bootstrap	0.00368
RiskMetrics ($\lambda = 0.94$)	0.00368
GARCH-Gaussian	0.00369

Several points are important.

First, **Historical Simulation and CAViaR-SAV have by far the widest average intervals**. For Historical Simulation, this is natural: the 1% quantile is estimated from a finite rolling sample of past returns, so tail estimation noise is substantial. For CAViaR, the wide intervals reflect the fact that the interval is not fully model-based; instead, the code uses a shifted empirical proxy. In both cases, the width indicates materially greater uncertainty around the estimated tail threshold.

Second, **the best-performing FHS models balance calibration and uncertainty reasonably well**. GJR-FHS has a smaller mean interval width than GARCH-FHS and GARCH-t-FHS, while also delivering the strongest backtest results. That combination is appealing: it suggests a forecast that is both well-calibrated and relatively stable. GARCH-FHS is slightly wider, but still much tighter than Historical Simulation or CAViaR and statistically very strong.

Third, **narrower intervals do not automatically mean better risk forecasts**. GARCH-Gaussian, RiskMetrics, and the package-based GARCHX benchmark all have relatively tight intervals, but their backtest performance is weaker than that of GJR-FHS and GARCH-FHS. This is an important practical lesson: a model can be confidently wrong. A narrow interval around a badly calibrated VaR forecast is not a virtue.

Fourth, **bootstrap methods produce tight intervals because the one-step construction is mechanically concentrated**, especially in this project where one-step bootstrap and FHS are already close. GARCH-t-Bootstrap and GARCH-Bootstrap therefore show comparatively small average widths. Their strong backtests indicate that these intervals are not merely artifacts of underreaction, but they should still be interpreted with the project's deterministic bootstrap seeding limitation in mind.

Fifth, **the package-based GARCHX(VIX) benchmark has a relatively narrow interval but weaker coverage**. This suggests that adding VIX improves neither tail calibration nor practical uncertainty characterization enough to beat the best empirical-tail models. Its forecast distribution is fairly concentrated, but that concentration is not translating into superior downside-risk reliability.

6.2 Practical interpretation of uncertainty

From a risk-management perspective, the lower and upper bounds are useful for more than presentation. They indicate how much confidence one should place in the estimated 1% threshold on a given day.

- A **wide interval** means the left-tail estimate is itself unstable; the reported VaR should be treated cautiously because a modest change in assumptions or sample composition could shift the tail threshold meaningfully.
- A **narrow interval** means the point estimate is precise within that model's own framework, but it does **not** guarantee good risk control unless the model is also well-calibrated in backtests.
- The most practically useful models are those that combine **good backtesting performance** with **moderate and interpretable interval width**. In the saved outputs, GJR-FHS and GARCH-FHS fit that description best.

Overall, the uncertainty analysis reinforces the main empirical ranking rather than overturning it. The strongest models are not those with the narrowest intervals in isolation, but those with intervals that are reasonably tight **and** accompanied by good unconditional coverage, independence, and conditional coverage performance.

7 Discussion

The main empirical lesson is that tail modeling matters more than merely changing the volatility recursion. The strongest models all combine dynamic conditional scale with an empirical description of the lower-tail innovation distribution. This is why the FHS variants dominate the simple Gaussian approaches.

Asymmetry helps, but only after the tail is handled properly. The difference between GJR-FHS and GARCH-FHS suggests that leverage effects add value, but the larger gain comes from empirical-tail calibration rather than from asymmetry alone.

The project also shows that a standard exogenous-volatility benchmark does not automatically improve VaR performance. VIX is economically meaningful, but in the saved outputs it does not translate into a better one-day 1% left-tail forecast than the best FHS models.

Finally, the uncertainty results add an important nuance. Broad intervals from Historical Simulation and CAViaR indicate real imprecision in tail estimation, while very tight intervals from weaker models such as GARCH-Gaussian show that numerical precision without calibration is not enough. The best models are those that are both statistically credible and reasonably stable.

8 Limitations

Several limitations should be stated clearly.

1. The project now spans multiple runners, so the final comparison is less tidy than it would be in a single integrated script.
2. The package-based GARCHX benchmark still uses a Gaussian VaR mapping, so its tail remains parametric even though the variance recursion is standard and defensible.
3. The CAViaR implementation is only a partial replication of the original paper: only SAV is implemented, optimization is limited, and the Dynamic Quantile test is absent.
4. The bootstrap implementation resets the random seed internally, making the saved rolling bootstrap effectively deterministic.
5. The conclusions are specific to one asset, one forecast horizon, and one VaR level. A broader study could compare multiple assets, horizons, and Expected Shortfall in addition to VaR.

9 Conclusion

This report updates the written project summary to match the current repository contents, presentation, plots, tables, and forecast files for one-day-ahead 1% VaR forecasting on QQQ daily log returns.

The saved evidence is consistent across tables, plots, and backtests. Empirical-tail GARCH models are the strongest methods in this project. GJR-FHS is the best overall model, with GARCH-FHS as a very close second. Student-*t* empirical-tail variants also perform strongly, and Historical Simulation remains a useful benchmark. By contrast, Gaussian GARCH and RiskMetrics underestimate downside risk, while the package-based GARCHX(VIX) benchmark does not outperform the best empirical-tail alternatives.

The new uncertainty section strengthens the interpretation of these results. VaR should not be read as a single deterministic number. The saved lower and upper bounds show that uncertainty differs substantially across models. Historical Simulation and CAViaR are much less precise; Gaussian-type models can be precise but poorly calibrated; and the most convincing practical models are those such as GJR-FHS and GARCH-FHS that combine strong backtesting with moderate, interpretable interval width.

The main conclusion is therefore unchanged but now more fully supported: for one-day-ahead 1% VaR on QQQ, the best-performing forecasts come from models that estimate volatility dynamically and calibrate the tail empirically rather than relying on a thin-tailed parametric quantile.

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